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An Average Flow Model for Determining Effects of Three-Dimensional Roughness on Partial Hydrodynamic Lubrication

A new approach is utilized to determine the effects of surface roughness on partially lubricated contacts. An average Reynolds equation for rough surfaces is defined in terms of pressure and shear flow factors, which are obtained by numerical flow simulation. Through the use of measured or numerically generated rough surfaces, any three dimensional roughness structure can be analyzed with this method. The average Reynolds equation is obtained for isotropic surfaces, and for surfaces with directional patterns. The flow factors for these surfaces are expressed as empirical relations in terms of h/σ and a surface characteristic γ defined as the ratio of x and y correlation lengths.

Introduction

The study of surface roughness effects in lubrication has gained an increasing attention with the introduction of stochastic concepts by Tzeng and Saibel [1].[1] Since then stochastic process theory has been utilized on various bearing performances. Tzeng and Saibel [1] utilized the stochastic approach for a slider bearing with one dimensional transverse roughness. Christensen and Tonder developed the stochastic Reynolds equation for transverse and longitudinal roughness and used this equation to analyze the hydrodynamic lubrication of slider and journal bearings, [2,3,4,5,6]. The effects of two sided-striated roughness on bearing load carrying capacity is obtained by Rhow and Elrod [7]. Chow and Cheng [8,9] extended the stochastic theory to Elastohydrodynamic (EHD) lubrication between two rollers. Cheng and Dyson [10] obtained a complete solution for the inlet half of an EHD contact between two circumferentially ground rough disks.

But most of these results are limited to two specific types of roughness structures; one dimensional ridges oriented either transversely or longitudinally. The extension to real three dimensional surface roughness is very difficult with the existing stochastic theory. There is a need for a new approach to study isotropic or nonisotropic three dimensional surface roughness.

This paper introduces a new method of deriving the average Reynolds equation through Flow Simulation. It is based on numerically solving the Reynolds equation on a model bearing with a randomly generated surface roughness and then deriving the average Reynolds equation from mean flow quantities. Thus by generating the surface roughness with known statistical properties (or alternatively, using the height readings of a real surface) any type of roughness structure can be analyzed. Another advantage of this method is that it can be extended to the partial lubrication regime ($h/\sigma < 3$) where the effect of roughness is most important.

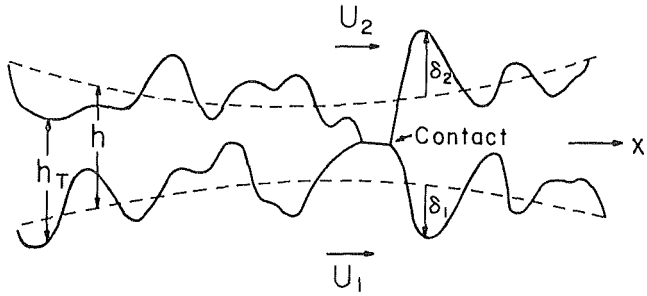


Fig. 1 Film thickness function

Film Thickness Function. The local film thickness h_T is defined to be of the form (see Fig. 1)

$$h_T = h + \delta_1 + \delta_2 \tag{1}$$

where h is the nominal film thickness (compliance) defined as the distance between the mean levels of the two surfaces. δ_1 and δ_2 are the random roughness amplitudes of the two surfaces measured from their mean levels. We assume δ_1 and δ_2 have a Gaussian distribution of heights with zero mean and standard deviations σ_1 and σ_2, respectively. (However the general theory is independent of the Gaussian

[1] Numbers in brackets designate References at end of paper.

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surface assumption.) The combined roughness $\delta = \delta_1 + \delta_2$ has a variance $\sigma^2 = \sigma_1^2 + \sigma_2^2$.

The ratio h/σ is an important parameter showing the effects of surface roughness. For $h/\sigma \gg 3$, the roughness effects are not important, and smooth film theory is sufficiently accurate. The roughness effects become important as $h/\sigma \to 3$. As h/σ is decreased further, asperities start interacting with each other and contacts form. This paper will be mainly concerned with $h/\sigma < 3$ which is termed as the "partial lubrication regime."

At contact points $h_T = 0$. The average gap $\bar{h}_T$ is then defined as

$$\bar{h}_T = \int_{-h}^{\infty} (h + \delta) f(\delta) d\delta \tag{2}$$

where δ is the combined roughness, and $f(\delta)$ is the probability density function of δ.

Average Reynolds Equation. For an isothermal, incompressible lubricant, the pressure in EHD contacts is governed by the Reynolds equation:

$$\frac{\partial}{\partial x}\left(\frac{h_T^3}{12\mu}\frac{\partial p}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{h_T^3}{12\mu}\frac{\partial p}{\partial y}\right) = \frac{U_1 + U_2}{2}\frac{\partial h_T}{\partial x} + \frac{\partial h_T}{\partial t} \tag{3}$$

We want to derive an average Reynolds equation which governs the average pressure in EHD contacts. To achieve this we analyze the expected flow in a rough bearing.

Consider a control volume with base area $\Delta x \Delta y$ and height h_T, as shown in Fig. 2. Assume $\Delta x \Delta y$ is large enough to include a large number of asperities, but still small compared to the size of the bearing.

The unit flow (volume flow per unit width) in the x and y directions are given by

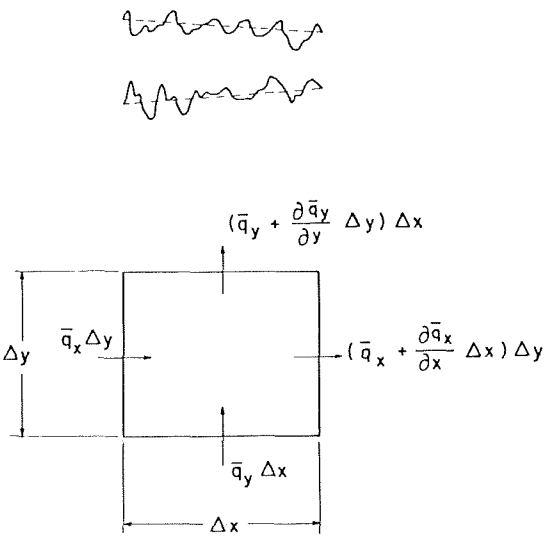


Fig. 2 Control volume for average flow

$$q_x = -\frac{h_T^3}{12\mu}\frac{\partial p}{\partial x} + \frac{U_1 + U_2}{2} h_T$$
$$q_y = -\frac{h_T^3}{12\mu}\frac{\partial p}{\partial y} \tag{4}$$

q_x and q_y are local flows and hence they are random functions. Consider now the mean (expected) unit flows entering into the control volume

$$\bar{q}_x = \frac{1}{\Delta y}\int_y^{y+\Delta y} q_x dy = \frac{1}{\Delta y}\int_y^{y+\Delta y}\left(-\frac{h_T^3}{12\mu}\frac{\partial p}{\partial x} + \frac{U_1 + U_2}{2} h_T\right) dy$$
$$\bar{q}_y = \frac{1}{\Delta x}\int_x^{x+\Delta x} q_y dx = \frac{1}{\Delta x}\int_x^{x+\Delta x}\left(-\frac{h_T^3}{12\mu}\frac{\partial p}{\partial y}\right) dx$$

Define pressure flow factors ϕ_x, ϕ_y, and shear flow factor ϕ_s such that the expected units flows are:

$$\bar{q}_x = -\phi_x \frac{h^3}{12\mu}\frac{\partial \bar{p}}{\partial x} + \left(\frac{U_1 + U_2}{2}\right)\bar{h}_T + \frac{U_1 - U_2}{2}\sigma\,\phi_s \tag{6}$$

$$\bar{q}_y = -\phi_y \frac{h^3}{12\mu}\frac{\partial \bar{p}}{\partial y} \tag{7}$$

where $\bar{p}$ is the mean (expected) pressure. ϕ_x and ϕ_y compare the average pressure flow in a rough bearing to that of a smooth bearing. The ϕ_s term represents the additional flow transport due to sliding in a rough bearing.

As assumed earlier, Δx and Δy are sufficiently large to include a large number of asperities. In this case although $\bar{q}_x$ and $\bar{q}_y$ are still random quantities, one expects them to have small variances. Hence ϕ_x, ϕ_y, and ϕ_s should also have small variances.

Performing a mean flow balance on the control volume.

$$\left(\bar{q}_x + \frac{\partial \bar{q}_x}{\partial x}\Delta x\right)\Delta y - \bar{q}_x \Delta y + \left(\bar{q}_y + \frac{\partial \bar{q}_y}{\partial y}\Delta y\right)\Delta x - \bar{q}_y \Delta x = -\Delta x \Delta y \frac{\partial \bar{h}_T}{\partial t} \tag{8}$$

or

$$\frac{\partial \bar{q}_x}{\partial x} + \frac{\partial \bar{q}_y}{\partial y} = -\frac{\partial \bar{h}_T}{\partial t} \tag{9}$$

Substituting equations (6) and (7), the average Reynolds equation is obtained:

$$\frac{\partial}{\partial x}\left(\phi_x \frac{h^3}{12\mu}\frac{\partial \bar{p}}{\partial x}\right) + \frac{\partial}{\partial y}\left(\phi_y \frac{h^3}{12\mu}\frac{\partial \bar{p}}{\partial y}\right) = \frac{U_1 + U_2}{2}\frac{\partial \bar{h}_T}{\partial x} + \frac{U_1 - U_2}{2}\sigma\frac{\partial \phi_s}{\partial x} + \frac{\partial \bar{h}_T}{\partial t} \tag{10}$$

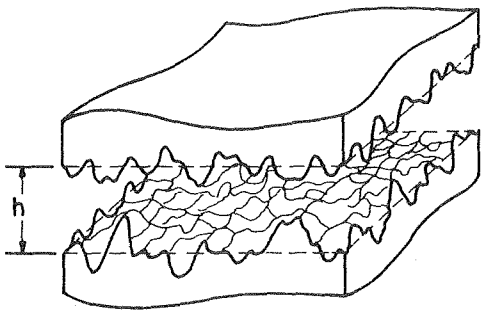
As $h/\sigma \to \infty$, this equation becomes the Reynolds equation for smooth surfaces provided the ϕ_x and ϕ_y satisfy

$$\phi_x, \phi_y \to 1 \text{ as } h/\sigma \to \infty \tag{11}$$

Nomenclature

h = nominal film thickness (compliance)
h_T = local film thickness
$\bar{h}_T$ = average gap
$H = h/\sigma$
p = hydrodynamic pressure
$\bar{p}$ = mean hydrodynamic pressure
q_x, q_y = unit oil flow in x and y directions
$\bar{q}_x, \bar{q}_y$ = mean (expected) unit flows
R_x, R_y = profile auto-correlation functions
U_1, U_2 = velocity (x direction) of surfaces
U = rolling velocity $U = \dfrac{U_1 + U_2}{2}$
γ = ratio of x and y correlation lengths $= \dfrac{\lambda_{0.5x}}{\lambda_{0.5y}}$
δ_1, δ_2 = roughness amplitudes of surfaces measured from their mean levels
δ = combined roughness $\delta = \delta_1 + \delta_2$
E = expectancy operator
λ = delay length
$\lambda_{0.5}$ = 0.5 correlation length of a profile
μ = absolute viscosity
σ_1, σ_2 = standard deviations of roughness functions δ_1 and δ_2
σ = standard deviations of combined roughness δ, $\sigma = \sqrt{\sigma_1^2 + \sigma_2^2}$
ϕ_x, ϕ_y = pressure flow factors
ϕ_s = shear flow factor

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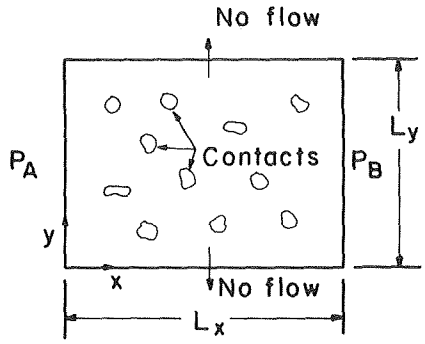


Fig. 3 The model problem for simulation

The flow factors ϕ_x, ϕ_y, ϕ_s will be obtained through flow simulation. This paper will only consider the pure rolling case $U_1 = U_2 = U$, and obtain ϕ_x, ϕ_y.

Derivation of φ_x, φ_y Through Simulation. Assume the bearing is approximated by a number of small rectangular bearings of area δA_i. The nominal film thickness h is assumed to be constant within each bearing. As before assume δA_i is small relative to the dimensions of the bearing, but large enough to include a large number of asperities.

For each rectangular bearing with a given roughness structure, the flow factors ϕ_x and ϕ_y can be calculated by applying an arbitrary pressure gradient on the boundaries, solving the pressure flow by numerical methods, and then comparing this flow to that of a similar smooth bearing. Hence, considering such bearings of different nominal gaps, ϕ_x and ϕ_y can be obtained as a function of h.

Thus to obtain ϕ_x, we consider the following model problem; (see Fig. 3)

$$\frac{\partial}{\partial x}\left(\frac{h_T^3}{12\mu}\frac{\partial p}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{h_T^3}{12\mu}\frac{\partial p}{\partial y}\right) = U\frac{\partial h_T}{\partial x} + \frac{\partial h_T}{\partial t} \tag{12}$$

$$h_T = h + \delta_1 + \delta_2 \; (h = \text{constant}) \tag{13}$$

with boundary conditions:

$$\begin{aligned} &1) \quad p = p_A \text{ at } x = 0 \\ &2) \quad p = p_B \text{ at } x = L_x \\ &3) \quad \frac{\partial p}{\partial y} = 0 \text{ at } y = 0, y = L_y \\ &4) \quad \text{No flow at contact points} \end{aligned} \tag{14}$$

Now consider the right-hand side of equation (12). Since h is constant;

$$U\frac{\partial h_T}{\partial x} + \frac{\partial h_T}{\partial t} = U\frac{\partial(\delta_1 + \delta_2)}{\partial x} + \frac{\partial(\delta_1 + \delta_2)}{\partial t} \tag{15}$$

Since the time dependence of δ_1, δ_2 comes from the movement of the two surfaces with velocity U, we can write

$$\delta_i = \delta_i(x - Ut, y) \quad i = 1,2 \tag{16}$$

Hence

$$\frac{\partial \delta_i}{\partial t} = -U\frac{\partial \delta_i}{\partial x} \quad i = 1,2 \tag{17}$$

Therefore the two terms on the RHS cancel and equation (12) becomes

$$\frac{\partial}{\partial x}\left(\frac{h_T^3}{12\mu}\frac{\partial p}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{h_T^3}{12\mu}\frac{\partial p}{\partial y}\right) = 0 \tag{18}$$

After generating δ_1 and δ_2 randomly with known statistical properties over the area of the bearing (or using measured real surfaces), the model problem (equations (18, 14)) is solved for pressure using finite difference methods. The average unit flow in the x direction $\bar{q}$ is computed using equation (5). ϕ_x is then calculated using equation (6). After cancellations ϕ_x becomes:

$$\phi_x = \frac{\dfrac{1}{L_y}\displaystyle\int_0^{L_y}\left(\frac{h_T^3}{12\mu}\frac{\partial p}{\partial x}\right)dy}{\dfrac{h^3}{12\mu}\dfrac{\partial \bar{p}}{\partial x}} \tag{19}$$

where

$$\frac{\partial \bar{p}}{\partial x} = \frac{p_B - p_A}{L_x} \tag{20}$$

ϕ_x thus obtained is dependent on the precise roughness geometry of the bearing. In order to obtain the expected ϕ_x, one should solve the same problem a number of times with different but statistically identical roughness functions, and take the average of the ϕ_x. Each of the ϕ_x values reported later in the paper are the average of about 10 different solutions.

The calculation of ϕ_y follows the same steps as the calculation of ϕ_x. In this case we apply an arbitrary pressure gradient in the y direction and calculate the expected unit flow $\bar{q}_y$, and compute ϕ_y from equation (7). As expected, the equations for calculating ϕ_y become exactly the same as the equations for ϕ_x. In fact if the directional properties of the roughness are the same in x and y directions, $\phi_x = \phi_y$. Since ϕ_x and ϕ_y depend on the directional properties of the roughness functions, the results will only be presented for ϕ_x. ϕ_y will then be equal to the ϕ_x value corresponding to the directional properties of the y profile.

ϕ_x is a function of h/σ and the statistical properties of the combined roughness δ. Among these properties, the frequency density of roughness heights and the directional properties of the surfaces are most important.

Since most engineering surfaces have (approximately) Gaussian distribution of heights, the results presented in this paper are for Gaussian surfaces.

To study surfaces with directional properties, we will use a surface characteristic γ defined by Kubo and Peklenik [11]. Let $\lambda_{0.5}$ be the length at which the auto-correlation function of a profile reduces to 50 percent of its initial value. γ is defined as the ratio of $\lambda_{0.5}$ lengths of the x and y profiles, i.e.

$$\gamma = \frac{\lambda_{0.5x}}{\lambda_{0.5y}} \tag{21}$$

γ can be visualized as the length-to-width ratio of a representative asperity. Purely transverse, isotropic and longitudinal roughness patterns correspond to $\gamma = 0, 1, \infty$, respectively.

The methods of generating isotropic and directional surfaces are outlined in the Appendix. The next sections present the simulation results for such surfaces.

Isotropic Surface Roughness. For an isotropic roughness structure ($\gamma = 1$), there are no preferred orientations (i.e., statistical

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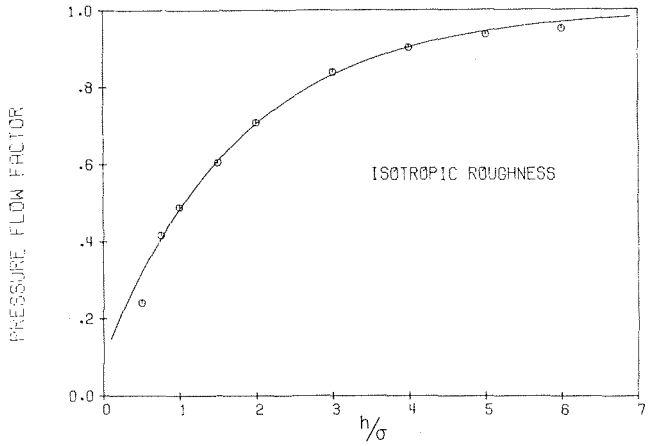


Fig. 4 The pressure flow factor ϕ_x or ϕ_y for isotropic surfaces

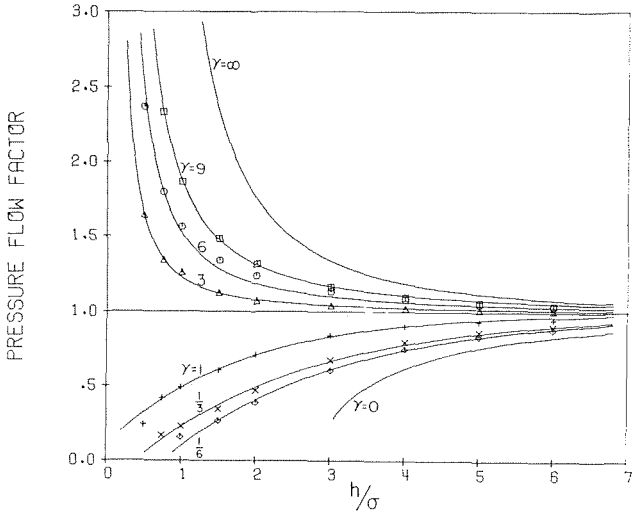


Fig. 5 Effect of directional properties of roughness on pressure flow factor ϕ_x. Curves for $\gamma = 0$ and $\gamma = \infty$ are obtained from stochastic theory for transverse and longitudional ridges

properties are constant along any direction). Therefore the pressure flow factors ϕ_x and ϕ_y should be equal.

Fig. 4 shows the pressure flow factor ϕ_x for isotropic roughness as a function of h/σ. Each data point is the average of about 10 simulation solutions obtained by using different, but statistically identical, rough surfaces. For $h/\sigma > 1$, the ϕ_x values obtained are closely clustered about the mean value, but as $h/\sigma \to 0$ the scatter in these values increase sharply. This is mainly due to the increased sensitivity of the flow factor on h/σ and on the directional properties of the roughness. A slight deviation from isotropy results in a large change in ϕ_x, thus the flow factors obtained from different surfaces show a large scatter.

The ϕ_x for isotropic roughness is close to 1 for $h/\sigma > 6$, which means that the average Reynolds equation is the same as that of smooth surfaces. This is an expected result since we know that longitudinal roughness enhances pressure flow ($\phi_x > 1$), while transverse roughness restricts the flow ($\phi_x < 1$). For an isotropic roughness structure, these two effects somehow "cancel" each other resulting in a flow that is the same as a smooth bearing.

However, in the mixed lubrication regime $h/\sigma < 3$, the asperities start interacting with each other and form barriers to flow, resulting in a sharp decrease in the pressure flow and hence in the pressure flow factor.

The ϕ_x versus h/σ curve is closely approximated by the equation

$$\phi_x = 1 - .90\, e^{-.56H} \tag{22}$$

where $H = h/\sigma$.

Surfaces With Directional Patterns. Many engineering surfaces have directional patterns resulting from different manufacturing processes or because of "running-in." These directional patterns are mostly in the longitudinal or transverse directions.

Pressure flow factors for such surfaces are obtained by generating rough surfaces with different γ ratios. The results are shown in Fig. 5. For comparisons, ϕ_x for one dimensional longitudinal ($\gamma = \infty$) and transverse ($\gamma = 0$) roughness, derived by stochastic process theory are also shown in the figure. Flow factors for these cases are obtained by comparing the stochastic Reynolds equations [3,6,8] with the average Reynolds equation defined in this paper. Comparison yields:

$$\phi_x = \frac{1}{h^3 E\left(\frac{1}{h_T{}^3}\right)} \text{ (Transverse)} \tag{23}$$

$$\phi_x = \frac{E(h_T{}^3)}{h^3} \text{ (Longitudinal)} \tag{24}$$

where E is the expectancy operator. Simulation results for purely transverse and longitudinal roughness agree well with these graphs.

For surfaces having larger correlation lengths in the longitudinal direction ($\gamma > 1$), the pressure flow factor ϕ_x is greater than 1. While larger correlation lengths in the transverse direction ($\gamma < 1$) yield ϕ_x smaller than 1. As expected, the flow factors all approach to 1 asymptotically as h/σ increases. But as h/σ decreases below 3, the flow

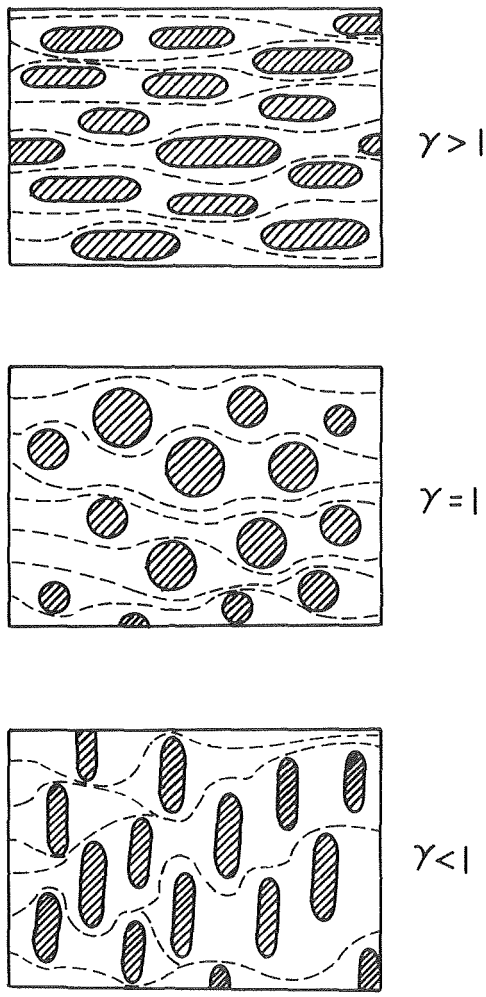


Fig. 6 Typical contact areas for longitudionally oriented ($\gamma > 1$), isotropic ($\gamma = 1$) and transversely oriented ($\gamma < 1$) surfaces

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Table 1 Coefficients of equations (25), (26)

γ	C	r	Range
1/9	1.48	0.42	H > 1
1/6	1.38	0.42	H > 1
1/3	1.18	0.42	H > .75
1	0.90	0.56	H > .5
3	0.225	1.5	H > .5
6	0.520	1.5	H > .5
9	0.870	1.5	H > .5

factors become very sensitive to the directional properties of the surface roughness.

The sensitivity of the flow factor on γ in the partial lubrication regime can be explained by a simple visual experiment. For a general three dimensional surface, contact areas can be modeled as ellipses with the mean ellipticity ratio being equal to γ. Fig. 6 shows typical contact area configurations for three different ranges of the ellipticity parameter γ.

Longitudinally oriented contact areas ($\gamma > 1$) offer little resistance to the pressure flow, permitting only a small side flow. Since the average gap in the valleys is larger than the compliance, the resulting flow is greater than a similar smooth bearing. A decrease in h/σ is accompanied by a large increase in the mean gap (relative to h/σ), and hence a large increase in the flow factor. A decrease in γ, however, results in smaller valley lengths and increased side flow, thereby decreasing the main flow and the flow factors.

When the surface is isotropic ($\gamma = 1$), the local side flow is of the same order of magnitude as the main flow. Hence the resistance to main flow increases because the flow has to "pass around" the contact areas. This increase in resistance offsets the effects of a larger mean gap, resulting in $\phi_x < 1$.

The side flow and hence the resistance to the main flow is increased further as the surface becomes transversely oriented ($\gamma < 1$). A decrease in γ and/or in h/σ increases this resistance, thereby reducing the ϕ_x.

At low values of h/σ, there is also a considerable difference in ϕ_x between purely longitudinal rough surfaces ($\gamma = \infty$), and surfaces with finite γ values. Hence the approximation of real surfaces by one dimensional ridges may introduce large errors.

The data shown in Fig. 5 are fitted into empirical relations of the form

$$\phi_x = 1 - CE^{-rH} \text{ for } \gamma \leq 1 \tag{25}$$

$$\phi_x = 1 + CH^{-r} \text{ for } \gamma > 1 \tag{26}$$

where $H = h/\sigma$. The constants C and r are given as functions of γ in Table 1.

For a surface roughness with a given γ value, ϕ_y is equal to the ϕ_x value corresponding to $1/\gamma$, i.e.,

$$\phi_y(H, \gamma) = \phi_x(H, 1/\gamma) \tag{27}$$

Conclusion

In the past decade, the stochastic process theory have been used to determine effects of surface roughness on hydrodynamic bearings. But the assumption of one dimensional roughness rendered this theory less useful especially for predicting performance and failure of EHD contacts in the partial lubrication regime. The research outlined in this paper was motivated by the need for a new approach that can be utilized to study any general three dimensional roughness.

The new approach is based on defining an average Reynolds equation in terms of pressure and shear flow factors, which are functions of surface roughness characteristics. These flow factors can be derived independently, from the mean flow quantities obtained from the numerical solution of a model bearing with a randomly generated or measured surface roughness. This method is more versatile than the earlier stochastic theories because it can be readily applied to any three-dimensional surface roughness structure. The only drawback of this method is the tedious effort required to obtain the flow factors through simulation. But once they are obtained, they can be utilized with the average Reynolds equation to determine roughness effects on any type of bearing.

The empirical flow factors obtained in this paper can be immediately applied for estimating the mean flow in static seals. The concept of average Reynolds equation can be also employed to solve for the mean hydrodynamic pressure in rolling contacts provided a reliable load-compliance relationship for the mean asperity pressure is available.

Acknowledgments

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References

1 Tzeng, S. T., and Saibel, E., "Surface Roughness Effect on Slider Bearing Lubrication," *ASLE Trans,* Vol. 10, 1967, p. 334

2 Christensen, H., "Stochastic Models for Hydrodynamic Lubrication of Rough Surfaces," *Proc. Instn. Mech. Engrs.*, Tribology Group 184, Part 1, Vol. 55, 1969–70, p. 1013

3 Christensen, H., and Tonder, K., "The Hydrodynamic Lubrication of Rough Bearing Surfaces of Finite Width," JOURNAL OF LUBRICATION TECHNOLOGY, TRANS. ASME Series F, Vol. 93, 1971, p. 324

4 Christensen, H., and Tonder, K., "The Hydrodynamic Lubrication of Rough Journal Bearings," JOURNAL OF LUBRICATION TECHNOLOGY, TRANS. ASME Series F, Apr. 1973, p. 166

5 Tonder, K., and Christensen, H., "Waviness and Roughness in Hydrodynamic Lubrication," *Proc. Instn. Mech. Engrs.*, Tribology Grp., Vol. 186, 1972, p. 807

6 Christensen, H., "A Theory of Mixed Lubrication," *Proc. Instn. Mech. Engrs.* Tribology Group, Vol. 186, 1972, p. 421

7 Rhow, S. K., and Elrod, H. G., "The Effects on Bearing Load Carrying Capacity of Two-Sided Striated Roughness," JOURNAL OF LUBRICATION TECHNOLOGY, TRANS. ASME Series F, Vol. 96, 1974, p. 554

8 Chow, L. S. H., and Cheng, H. S., "Influence of Surface Roughness and Waviness on Film Thickness and Pressure Distribution in EHD Contacts," NASA Contractor Report CR-2670 1976

9 Chow, L. S. H., and Cheng, H. S., "The Effect of Surface Roughness on Average Film Thickness Between Lubricated Rollers," JOURNAL OF LUBRICATION TECHNOLOGY, TRANS. ASME, Series F, Vol. 98, 1976 p. 117

10 Cheng, H. S., and Dyson, A., "EHD Lubrication of Circumferentially Ground Rough Disks," ASLE Paper No: 76-LC-1A-2 1976

11 Peklenik, J., "New Developments in Surface Characterization and Measurement by Means of Random Process Analysis," *Proc. Instn. Mech. Engrs.*, Vol. 182, Part 3K; 1967–68 p. 108

12 Patir, N., "A Numerical Procedure for Random Generation of Rough Surfaces," to be published in *WEAR.*

Appendix

Generating Random Roughness. The use of randomly generated surface roughness is an important tool for the analysis used in this paper. Random roughness generation is partly necessitated by the unavailability of accurate relocation techniques in the measurement of roughness. Furthermore a parameteric study of roughness requires surfaces with known statistical parameters, and it is much more convenient to generate them numerically, rather than measure mechanically produced rough surfaces. Nevertheless, the use of measured roughness would be justified if the properties of a specific surface is desired.

We now consider generating a matrix of roughness heights $[\delta_{ij}]$ with prescribed statistical properties. δ_{ij} corresponds to the roughness amplitude at $x = i\Delta l$, $y = j\Delta l$, where Δl is the "sampling interval." We assume the roughness height distribution is Gaussian, and we want the roughness to have different correlation lengths in x and y directions.

Auto-correlation functions of the x and y profiles are defined as:

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$$R_x(\lambda) = E(\delta_{i,j} \cdot \delta_{i+\lambda^*,j})$$

$$R_y(\lambda) = E(\delta_{i,j} \cdot \delta_{i,j+\lambda^*})$$

Hence the auto-correlation function shows the correlation between two points separated by a distance $\lambda = \lambda^* \Delta l$. The 0.5 correlation length $\lambda_{0.5}$ is defined as the length at which the profile correlation function reduces to 50 percent of its value at $\lambda = 0$.

For convenience, we assume the x and y profiles have linear auto-correlation functions. For small delay lengths λ, this is a reasonable approximation to real surfaces.

It is possible to generate surfaces with a given auto-correlation function using linear transformations on random matrices. We first generate a matrix $[z_{ij}]$ whose components are independent, identically distributed Gaussian random numbers with zero mean and unit standard deviation. The roughness heights are then obtained through the linear transformation:

$$\delta_{ij} = \sum_{k=1}^{n} \sum_{l=1}^{m} z_{i+k,j+l}$$

This transformation will result in x and y profile correlation functions

$$R_x(\lambda) = \begin{cases} (n - \lambda^*)m & 0 \le \lambda^* < n \\ 0 & n \le \lambda^* \end{cases}$$

$$R_y(\lambda) = \begin{cases} (m - \lambda^*)n & 0 \le \lambda^* < m \\ 0 & m \le \lambda^* \end{cases}$$

where $\lambda^* = \dfrac{\lambda}{\Delta l}$.

The parameter γ is then determined by:

$$\gamma = \frac{\lambda_{0.5x}}{\lambda_{0.5y}} = \frac{\frac{n\Delta l}{2}}{\frac{m\Delta l}{2}} = \frac{n}{m}$$

Hence the generated roughness δ will have a Gaussian height distribution with zero mean, variance = mn, and $\gamma = \dfrac{n}{m}$.

For additional information on surface roughness generation see [12].

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